

Benchmarking Literature Retrieval for a Model Organism: A *Dictyostelium* Case Study

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Abstract. Biological literature retrieval systems are often developed and evaluated using broad biomedical corpora and general-purpose search tasks. However, many curated knowledge bases operate in narrower model-organism domains, where the literature is sparse and terminology is organism-specific. We introduce a retrieval benchmark from dictyBase for *Dictyostelium*, a model organism in cell and developmental biology. The benchmark consists of curator-authored biological claims used as declarative retrieval queries, with their inline-cited PubMed articles as relevance judgments, together with structured gene annotations. Using this benchmark, we study three factors in niche biological retrieval: cross-encoder reranking, gene-aware query expansion, and abstract-only versus full-text retrieval. We report that reranking and gene-aware query expansion improve retrieval selectively: reranking gains are strongly model-dependent, whereas curated annotations help clarify compact claims by reducing vocabulary mismatch. Where full text is available, adding body chunks to the retrieval corpus substantially improves candidate recall and top-rank performance for claims whose cited abstracts lack sufficient evidence, although these cases remain harder than claims supported by abstracts. Data and code are publicly available at <https://github.com/fulaibaowang/dictycite>, and the benchmark dataset is archived on Zenodo.

Keywords: Biomedical literature retrieval · Model organism · Full-text retrieval · Query expansion · *Dictyostelium*.

1 Introduction

Biological literature retrieval systems are usually developed and benchmarked on broad biomedical corpora and general search tasks. However, a large amount of biological knowledge is maintained in narrower model-organism communities, such as *D. discoideum*, *D. melanogaster*, *C. elegans*. In these settings, the literature is smaller, terminology is more organism-specific, and relevant evidence

may depend on gene names, aliases, product annotations, and phenotype terms rather than on broadly shared biomedical vocabulary. Retrieval behavior in such domains may therefore differ from what is observed in large-scale biomedical question answering or general literature search.

Such differences raise a practical question for scientific retrieval systems: how well do standard retrieval pipelines work when a detailed biological claim is used as a query to retrieve relevant literature? Two commonly used strategies that we evaluate in this setting are semantic reranking, which reorders initially retrieved articles using a more selective relevance model, and query expansion, which enriches the query with related terms.

Many model-organism databases maintain gene-centered pages that combine structured annotations, textual summaries, and links to supporting literature [10, 15]. These resources rely heavily on expert biocuration. Among these resources, dictyBase covers *Dictyostelium*, a social amoeba widely used in biology, including studies of chemotaxis, cell motility, phagocytosis, and multicellular development. dictyBase is especially useful for studying retrieval in this setting because its gene summary pages contain curator-authored biological claims with inline citations. We use each claim as a declarative retrieval query and its cited article as an article-level relevance judgment. These claims may describe gene products, functions, expression patterns, phenotypes, regulatory interactions, or biological processes, often with additional developmental, experimental, or temporal context.

In this paper, we construct a citation-derived retrieval benchmark from dictyBase [6] and use it to study literature retrieval for *Dictyostelium*. We examine three questions: whether reranking adds value beyond strong keyword-based retrieval, whether curated annotations improve retrieval for compact claims, and how access to full-text evidence changes retrieval when abstracts are insufficient.

Our principal contribution is not a new retrieval architecture, but a benchmark and analysis of retrieval behavior in a focused model-organism domain. The results show that reranking gains depend strongly on model fit, that trusted gene annotations can reduce vocabulary mismatch, and that full-text chunks improve recall and final ranking when abstracts omit detailed evidence, while not fully closing the gap to claims whose evidence is already present in abstracts.

2 Related Work

Biomedical literature retrieval has been studied through benchmarks and shared tasks such as BioASQ [20] and the TREC BioGen track [9], including settings in which expert-authored biomedical questions are used to retrieve relevant literature from broad corpora. Closer to our query formulation, SciFact uses declarative scientific claims to retrieve evidence-bearing abstracts, but as part of a verification task that also classifies support or refutation [21]. We likewise use a declarative claim as the query, but derive claims and retrieval targets from naturally occurring curator claim-citation links in a model-organism knowledge

base, with an organism-scoped corpus and structured gene annotations available as query-side signals.

Modern retrieval systems commonly use two-stage pipelines that combine efficient first-stage retrieval with more expensive reranking. Lexical retrieval [18], dense retrieval, reciprocal rank fusion [5], cross-encoder reranking [3], and LLM-based rerankers are widely used components. Biomedical rerankers such as Med-CPT further demonstrate the value of domain adaptation for PubMed-scale retrieval [11]. Cross-domain evaluations such as BEIR show substantial variation in retrieval effectiveness across datasets, while BM25 remains a robust baseline [19]. We use these standard components to examine how reranker capacity and domain fit affect retrieval in a focused model-organism corpus, where a small number of organism-specific lexical cues may carry substantial relevance signal.

Query expansion and full-text retrieval are established strategies in biomedical search. Prior work has used controlled vocabularies, MeSH terms, gene synonyms, and concept-based expansion [13], and more recently language models that generate expansion terms or hypothetical documents [7, 22], to reduce vocabulary mismatch. Full-text and passage-level retrieval can additionally recover evidence that is absent from abstracts, although indexing finer-grained text introduces a larger and noisier candidate space [12]. We study these strategies using curated dictyBase annotations for query expansion and by comparing abstract-only retrieval with retrieval over chunked full text.

3 Dataset

We constructed a *Dictyostelium* literature retrieval dataset from dictyBase gene summary pages. Each retrieval instance is derived from a curator-authored biological claim and its inline literature citation: the claim becomes the query, and the cited PubMed-indexed articles are used as article-level relevance judgments. The dataset contains 1,656 curator-claim queries from 862 unique *Dictyostelium* genes, linked to 1,289 unique PubMed articles and evaluated against a 20,447-article Europe PMC abstract corpus. In addition to the query–article relevance pairs, the dataset includes structured gene annotations for gene-aware query expansion and LLM-assisted evidence labels for stratified analysis of whether the cited abstract contains sufficient evidence for the curator claim. The dataset is available on Zenodo under DOI 10.5281/zenodo.20308282.

Retrieval corpus. We assembled an organism-scoped retrieval corpus by collecting the full set of articles related to *Dictyostelium* from Europe PMC. We retained all records with available abstracts, yielding 20,447 articles after removing formatting tags and normalizing whitespace. All abstract-only retrieval and reranking experiments use this same corpus.

Curator-derived queries and relevance judgments. Queries were derived from curator notes on dictyBase gene summary pages. These notes consist of compact biological claims supported by inline citations, as illustrated in Fig. 1. For example, the claim that during normal development on a moist surface, *pkaC* mRNA

A. dictyBase source page

General Information	
Gene Name	<i>pkaC</i>
Name Description	pkaC = Protein Kinase A, Catalytic subunit
Alternative Gene Names	PKA, pkacat, DdPK3, DdPK2, PKA-C, PK2, PK3
Gene ID	DDB_G0283907
Gene Product	cAMP-dependent protein kinase A, catalytic subunit
Alternative Protein Names	cAK
Description	AGC group, PKA family protein kinase; protein serine/threonine kinase; regulates spore cell differentiation
Curator Notes	<i>pkaC</i> encodes the catalytic subunit of the cAMP dependent protein kinase PKA (Mann <i>et al.</i> 1992, Anjard <i>et al.</i> 1993). In the absence of cAMP, PKA (<i>pkaC</i> , <i>pkaR</i>) exists as an inactive complex of the catalytic and regulatory subunits. cAMP binds cooperatively to two sites on PkaR which leads to the release of active PkaC. This protein kinase plays multiple roles throughout development (Harwood <i>et al.</i> 1992, Mann <i>et al.</i> 1997, Loomis <i>et al.</i> 1998, Loomis <i>et al.</i> 2015). During normal development on a moist surface, mRNA from <i>pkaC</i> increases smoothly from an initially low level to reach a peak at 20 hours (Rosengarten <i>et al.</i> 2015). When cells are developed in suspension with addition of 50 nM cAMP every 6 minutes, the level of <i>pkaC</i> mRNA stays low

B. Extracted retrieval instance

Query = extracted curator claim

During normal development on a moist surface, mRNA from *pkaC* increases smoothly from an initially low level to reach a peak at 20 hours

Cited gold article = inline citation

Rosengarten *et al.* 2015
PMID: 25887420
Title: Leaps and lulls in the developmental transcriptome of *Dictyostelium discoideum*

Auxiliary annotations

Annotations for query expansion Pkac: pkacat, DdPK3; cAMP-dependent protein kinase A, catalytic subunit

LLM-assisted evidence label for this query–article pair abstract_insufficient

Fig. 1. Construction of a retrieval instance from a dictyBase gene page. (A) The source page for gene *pkaC* (http://dictybase.org/gene/DDB_G0283907) contains structured gene annotations and curator-authored notes with inline citations. (B) A curator-authored claim is extracted as the query, and its inline citation is mapped to a PubMed-indexed article used as the cited relevance judgment. Auxiliary fields drive later experiments: gene annotations for query expansion, and the LLM-assisted evidence label for the full-text analysis.

increases from an initially low level to a peak at 20 hours is extracted as a retrieval query, and the cited Rosengarten *et al.* article is used as the reference article for evaluation. Because dictyBase uses internal publication identifiers, we first mapped cited publications to PubMed IDs. Claims were excluded when their citations could not be mapped to PubMed or when the cited articles were absent from the Europe PMC abstract corpus, removing about 2% of the claims in total. The remaining claims were cleaned and deduplicated. Near-duplicate or semantically similar claims were grouped within each gene, and one representative claim was retained as the final query.

Gene annotations for query expansion. For each gene page, dictyBase provides structured annotations that connect a canonical gene name to alternative gene names, protein names, and product descriptions. These annotations are useful because relevant papers do not always use the canonical gene name. For example, *pkaC* (Fig. 1A) encodes PKA-C, the catalytic subunit of cAMP-dependent protein kinase A, where cAMP stands for cyclic adenosine monophosphate. A paper may therefore refer to the gene, the protein, the kinase activity, or cAMP without using *pkaC* itself. We retain these fields for query expansion only, not as

part of the original query. They are used only in the gene-aware query-expansion experiments described in Section 4.2, where these annotations can be appended to the query when the relevant gene mention is unambiguously resolved. This expansion setting is available for 563 queries.

Evidence labels and full-text subset. Curator citations define article-level relevance, but they do not guarantee that the cited abstract contains the specific evidence needed to support the full curator claim. For example, in the *pkaC* instance shown in Fig. 1B, the cited abstract describes the study at a broad level, but not the specific expression pattern stated in the curator claim. This gap creates an important distinction for retrieval evaluation: for some claims, the relevant evidence is already present in the abstract, whereas others may require evidence from the article body. We assigned an LLM-assisted evidence-level label to each claim–article pair to indicate whether the cited abstract: (i) supports the detailed claim—explicitly supports both the main assertion and its relevant specific details; (ii) supports only the core statement—the main assertion (for example, that a gene is involved in a process) but not the accompanying details (such as a specific developmental stage, condition, or magnitude); or (iii) is insufficient, suggesting that the abstract does not explicitly support the relevant relationship. The third label does not mark the citation as irrelevant; rather, it identifies cases where the supporting evidence may lie outside the abstract. The labeling prompt is released with the dataset and code. These labels define the strata for evaluating whether full-text retrieval helps at different levels of abstract evidence. Labels attach to claim–article pairs: because one claim can cite multiple articles, the 1,656 claims comprise 2,028 claim–article pairs (1.22 per claim), of which 41.6% support the detailed claim, 37.6% support the core claim, and 20.8% are insufficient. A claim enters every stratum whose label its cited articles carry: a claim citing three articles, two labelled *core* and one *insufficient*, contributes three pairs, two claim–stratum entries, and one distinct claim. We construct a full-text corpus for articles with accessible PDFs (1,124 cited articles); within a stratum, a claim is has-PDF if at least one of its articles carrying that label has full text, and no-PDF otherwise. Table 1 reports the composition.

Table 1. Benchmark composition by abstract-evidence stratum. Strata overlap because a claim enters each stratum represented among its cited articles. For claim–stratum entries, has-PDF means at least one cited article carrying that label has full text; for distinct claims, at least one cited article has full text.

Evidence stratum	All	has-PDF	no-PDF
<i>abstract supports detail</i>	816	707	109
<i>abstract supports core</i>	654	585	69
<i>abstract insufficient</i>	387	350	37
Claim–stratum entries	1,857	1,642	215
Distinct claims	1,656	1,480	176

4 Retrieval pipeline and variants

This section describes the retrieval pipeline and the query and corpus variants evaluated in the experiments.

4.1 Two-stage retrieval pipeline

Our pipeline follows a two-stage retrieval-and-reranking design (Fig. 2). Given a query, Stage 1 retrieves candidate articles from the abstract corpus using two complementary methods: BM25, which emphasizes overlap with the query’s exact terms, and dense retrieval, which ranks articles by semantic similarity between learned query and article representations. Reciprocal Rank Fusion (RRF) merges the two ranked lists, favoring articles ranked highly by either method, to form a single candidate pool.

In Stage 2, a cross-encoder reranks this pool using a more computationally expensive query–article relevance model. The final output combines the cross-encoder ranking with the original retrieval-fusion ranking through a second RRF step, retaining a small amount of first-stage signal. Stage 1 is designed to achieve high recall at large candidate depths, whereas Stage 2 aims to place cited articles near the top of the final ranking.

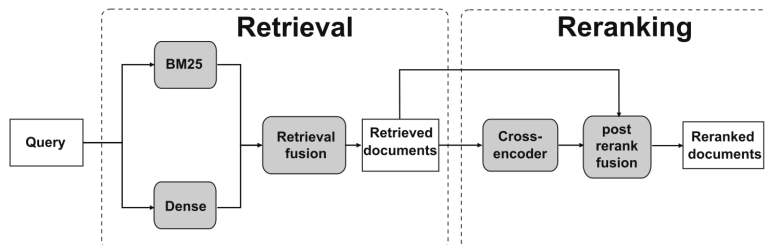


Fig. 2. Two-stage retrieval and reranking pipeline. Stage 1 fuses BM25 and dense retrieval with RRF into a recall-oriented candidate pool; Stage 2 reranks it with a cross-encoder and merges that order with the Stage-1 fusion through a second RRF.

4.2 Gene-aware query expansion

We use structured gene annotations from dictyBase to test whether gene-aware query expansion improves retrieval and downstream reranking. For each curator claim, we detect gene mentions using dictyBase gene IDs, canonical names, and synonyms, while filtering short, generic, or highly ambiguous aliases. We compare two expanded query variants with the original claim: (i) + synonyms, which appends the canonical gene name and selected synonyms, and (ii) + synonyms & products, which additionally appends a short product description. These nested variants represent progressively richer curated context, from alternative gene names to additional functional description. Gene IDs are used only for matching and are not appended to the query. The same expanded query text is used for both retrieval and reranking.

4.3 Chunked full-text corpus variant

To test whether evidence beyond abstracts improves retrieval, we compare the abstract-only corpus with a chunked full-text variant, where each article is split into smaller text segments, or chunks, that can be retrieved separately. In the abstract-only setting, each document is one PubMed abstract; in the full-text setting, each article is represented by an abstract chunk and body-text chunks. Body text is segmented by a greedy paragraph-based chunker (target 1,500 characters, hard cap 1,800, 200-character overlap, and a 300-character minimum; figure and table captions form separate chunks), giving on average about 24 body chunks per article across the 3,556 articles with accessible full text. The same retrieval, fusion, reranking, and post-rerank fusion steps are then applied without changing the pipeline parameters.

Because an article is split into several chunks, the same article can appear multiple times in a chunk-level ranking, while our relevance judgments are defined per article. We therefore collapse the chunk ranking back to articles: each article takes the rank of its highest-ranked chunk. The resulting article-level ranking lets us compare the abstract-only and chunked corpora on the same article-level relevance judgments.

5 Experimental setup

We evaluate all systems under fixed retrieval depths, fusion settings, and significance-testing procedures. This section defines the metrics and compared systems; implementation details are provided in the Appendix.

5.1 Metrics, candidate depths, and significance testing

We assess first-stage retrieval with Recall@K, which measures whether a cited article appears among the top K retrieved candidates, and the final ranking with MRR@K, which gives greater weight to placing the first cited article near the top. Because our judgments are derived from inline citations rather than exhaustive relevance assessments, MRR@10 is our primary final-ranking metric and should be interpreted as measuring retrieval of known cited articles, not retrieval of all relevant literature. In stratified comparisons, each claim is scored once per evidence stratum against only its cited articles carrying that label, so results are averaged over claim-stratum entries.

Pipeline parameters were fixed before the dictyBase experiments and were not tuned on this dataset. We retrieve 5,000 candidates per retriever, fuse the BM25 and dense lists, and pass the top 1,000 fused candidates to the reranker as a fixed, recall-oriented candidate depth.

For significance testing, we use paired bootstrap resampling over queries (within a stratum for stratified comparisons) with $B = 10,000$ samples. We report 95% percentile confidence intervals for per-query MRR@10 differences against the relevant baseline.

5.2 Systems compared

We compare BM25, BM25+Dense retrieval fusion without reranking, four cross-encoder rerankers, and one external listwise reference system. All cross-encoder rerankers are applied to the same BM25+Dense RRF candidate pool (Fig. 2). The rerankers are MiniLM-L12 (cross-encoder/ms-marco-MiniLM-L12-v2) [23], BGE-m3 (BAAI/bge-reranker-v2-m3) [4], MedCPT (ncbi/MedCPT-Cross-Encoder) [11], and BGE-Gemma (BAAI/bge-reranker-v2-gemma) [2], representing small general-purpose, strong multilingual, PubMed-trained biomedical, and large LLM-based CE reranking settings.

We also include Ragnarok BM25 + RankZephyr as an external reference [16, 17]. Ragnarok is an independently maintained retrieval-and-listwise-reranking pipeline using RankZephyr, a 7B listwise LLM reranker, and is intended as a contextual reference.

6 Results

We organize the results around three questions: whether cross-encoder reranking adds value beyond first-stage retrieval baselines, whether curated gene annotations help compact claims, and whether full-text chunks improve retrieval when abstracts lack sufficient evidence. The aggregate results are reported in Table 2 and Figs. 3–4; we use short query-level examples within each subsection to illustrate the mechanisms behind these results.

6.1 Reranking gains are model-dependent

On the 1,656-query dataset, BM25 is a strong baseline for this retrieval task (MRR@10 = 0.593; Table 2). BM25+Dense fusion is used as a recall-oriented candidate generator, but scores below BM25 at MRR@10, indicating that higher candidate recall does not automatically translate into better top-ranked results. Cross-encoder reranking is also not uniformly beneficial: the lightweight MiniLM-L12 reranker hurts performance, and BGE-m3 gives only a small, non-significant gain. Clear improvements appear for the biomedical MedCPT reranker and the high-capacity BGE-Gemma model, which improve MRR@10 by +5.7 and +11.4 percentage points, respectively. The external Ragnarok BM25+RankZephyr reference remains below BM25, further suggesting that gains depend on reranker fit.

The per-query analysis shows the same pattern (Fig. S1): MiniLM-L12 hurts more queries than it helps, BGE-m3 is nearly balanced, and MedCPT and BGE-Gemma shift more queries toward positive deltas. A concrete failure case is the query “*DymA* localizes to the phagosome and the cleavage furrow.” BM25 ranked the cited article second, whereas MiniLM-L12 moved it down to 60th. The article ranked first by MiniLM-L12 matched broad location terms, such as cleavage furrow and phagocytic cup, but lacked the *DymA*-specific signal. Thus, reranking condensed biological claims is useful only when the reranker preserves the task-relevant lexical signal.

Table 2. Ranker comparison on all 1,656 claims. Model size is the approximate parameter count for neural rerankers; “—” marks systems without one. Δ vs. BM25 is the MRR@10 difference from the BM25 baseline in percentage points, with brackets showing 95% paired-bootstrap confidence intervals.

System	Description	Model size	MRR@10	Δ vs. BM25
BM25	Lexical baseline	—	0.593	—
BM25+Dense	Lexical+dense RRF	—	0.568	−2.6 [−4.1, −1.0]
MiniLM-L12	Lightweight MS MARCO CE	33M	0.541	−5.2 [−7.0, −3.4]
BGE-m3	Strong multilingual CE	0.6B	0.605	+1.2 [−0.5, +2.8]
MedCPT	PubMed-trained biomedical CE	~110M	0.650	+5.7 [+4.2, +7.2]
BGE-Gemma	LLM-based CE reranker	3B	0.707	+11.4 [+9.8, +13.0]
Ragnorok	External BM25+RankZephyr	7B	0.566	−2.7 [−4.8, −0.8]

6.2 Gene-aware query expansion improves retrieval and benefits BGE-m3 and MedCPT

We next evaluate gene-aware query expansion on the 563-query subset for which dictyBase synonym and product fields are available. We compare the original query (the curator claim) with two expanded variants: adding gene synonyms, and adding both synonyms and product descriptions.

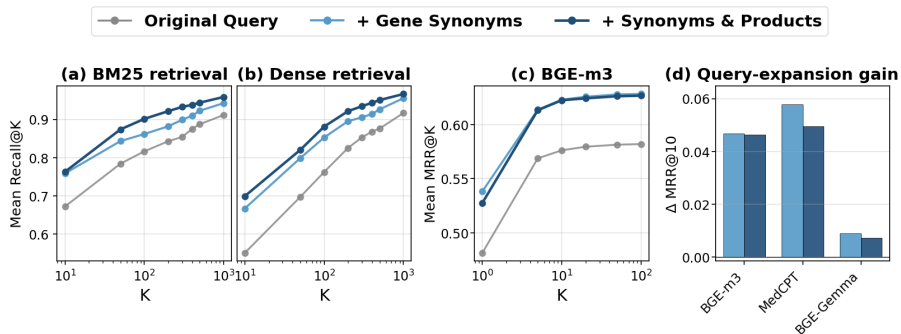


Fig. 3. Effect of gene-aware query expansion. Three query variants are compared: the original claim and the two gene-expansion variants (+synonyms, +synonyms & products; Section 4.2). (a) BM25 Recall@K, (b) dense Recall@K, and (c) mean MRR@K after BGE-m3 reranking, each plotted against the cutoff rank K on the x-axis. (d) Change in MRR@10 relative to the original-query baseline for BGE-m3, MedCPT, and BGE-Gemma; the two bars per reranker are the two expansion variants, and positive values indicate that expansion improved the final ranking. Paired-bootstrap confidence intervals for the MRR@10 deltas are reported in Table S1.

At the retrieval stage, expansion substantially improves recall for both BM25 and dense retrieval (Fig. 3a–b). For the reranking analysis, we fix the candidate pool to the one retrieved with the deepest expansion (+ synonyms & products), and vary only the query text passed to the reranker, isolating the query-formulation effect from the recall improvement above. Expanded reranker queries improve MRR@10 for BGE-m3 and MedCPT under both expansion variants, with all four confidence intervals above zero (Fig. 3c–d, Table S1). By contrast,

BGE-Gemma already performs strongly with the original query, and expansion adds less than one point (CIs cross zero).

A clear example is the query “*DetA* is involved in cell-type specification during development.” The cited article uses the synonym *DET1* rather than *DetA*. In the controlled reranking setting, adding this curated synonym moved the cited article from 39th to first under BGE-m3. The synonym supplies the surface form the article actually uses, closing the vocabulary gap between the compact curator wording and the article text. The effect is much smaller for BGE-Gemma, indicating diminishing returns when the reranker already recovers much of the relevant signal from the original query.

6.3 Full-text chunks benefit claims whose evidence is missing from the abstracts

The previous experiments used the abstract corpus. This setting is limiting for claims where the cited abstracts do not contain sufficient evidence. We therefore evaluate the same retrieval-and-reranking pipeline on the chunked full-text corpus variant (Section 4.3).

We stratify claims by the evidence-level labels introduced in the Dataset section: *abstract supports detail*, *abstract supports core*, and *abstract insufficient*, and report the main comparison on the has-PDF subset of each stratum, in which at least one cited article carrying that label has chunked full text (Fig. 4).

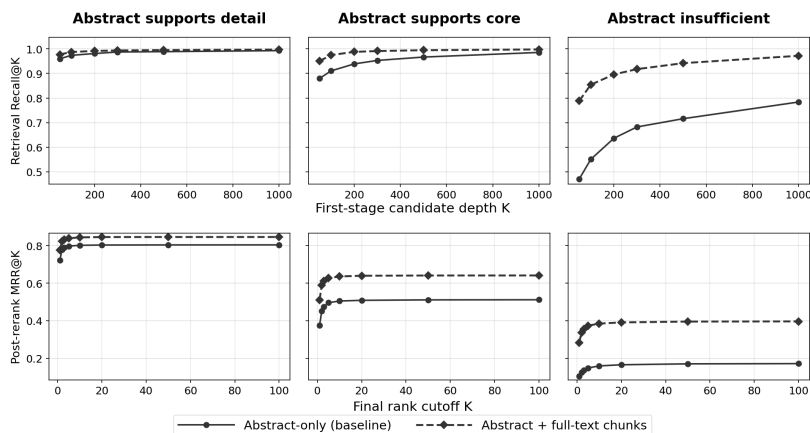


Fig. 4. Retrieval and reranking by abstract-evidence level on the has-PDF subset. The upper row shows first-stage Recall@K from BM25+Dense RRF retrieval. The lower row shows post-rerank MRR@K using BGE-m3. Columns correspond to *abstract supports detail* (n=707), *abstract supports core* (n=585), and *abstract insufficient* (n=350). Solid lines show the abstract-only baseline; dashed lines show the corpus augmented with full-text chunks.

The main effect of full-text chunks is at the retrieval stage. For abstract-insufficient claims, first-stage Recall@1000 rises from 0.78 to 0.97 (Fig. 4, upper panel), nearly closing the gap to the two abstract-supported groups. The other

two groups were already close to the ceiling under the abstract-only corpus and therefore show only small recall gains. Thus, when the cited abstract lacks the required evidence, adding full-text chunks substantially improves whether the cited PMID appears in the candidate pool.

The ranking stage also improves, although the evidence-level gap remains (Fig. 4, lower panel). With BGE-m3, post-rerank MRR@10 on abstract-insufficient claims more than doubles from 0.16 to 0.39, yet stays well below the two abstract-supported groups under the same chunked setting. The gain is significant across all three strata (95% paired-bootstrap CIs above zero; Table S2).

At the per-query level, chunked full-text retrieval improved many more abstract-insufficient claims than it hurt (46.9% improved, 8.0% decreased). A typical case is the query “*gefE* null mutants are less sensitive to DIF than wild type.” In the abstract-only corpus, the cited article ranked 54th because the abstract gives broad developmental context but omits the specific DIF-sensitivity evidence. With full-text chunks, the cited article ranked first because the relevant evidence appears in the article body.

This benefit is concentrated where the cited articles carrying the label gain full-text representation. For the complementary no-PDF entries, whose cited articles carrying that label contribute no body chunks, adding chunks leaves Recall@1000 unchanged or lower and significantly reduces post-rerank MRR@10 in all three strata (Table S2). These relevant articles gain no additional retrieval units but must compete against the added chunks. We return to this coverage trade-off in the Discussion.

7 Discussion

We built and released a benchmark for *Dictyostelium* literature retrieval that reflects a realistic form of a niche biological search: finding supporting citations for compact, gene-centered biological claims. The benchmark’s value is a combination of a new dataset and a test case for retrieval behavior that can differ from many open-domain QA settings. In this model-organism setting, gene symbols, pathway terms, phenotypes, and other organism-specific terms provide highly informative lexical cues. These cues help explain why BM25 remains strong and why reranking helps only when the model preserves these signals while adding useful semantic discrimination.

This finding has a practical implication for standard two-stage retrieval systems: cross-encoder reranking should not be treated as a default upgrade to the retrieval pipeline. In niche biological search, model choice, latency, and domain fit should be evaluated together. A reranking stage that adds substantial computation is useful only if it improves over a strong lexical baseline on the target retrieval task.

Gene-aware query expansion provides a complementary, low-overhead way to improve the pipeline. In our controlled reranking analysis, we fix the candidate pool and vary only the reranker query, showing that expansion helps not only by

retrieving more candidates but also by supplying gene vocabulary missing from compact claims. Unlike free-text or generated expansions, trusted entity-linked annotations are less likely to introduce noisy terms to which a cross-encoder may be sensitive. More broadly, trusted gene annotations paired with a smaller domain-suitable reranker may offer a more efficient operating point than relying only on larger LLM-based rerankers.

The full-text experiments show that abstracts are efficient but incomplete retrieval units. Body chunks help most when cited abstracts are insufficient, although the remaining MRR gap shows that recovering body evidence does not eliminate the ranking challenge. Incomplete coverage also creates an asymmetry: an article without full text contributes a single retrieval unit while competing against many chunks from articles that have it. Naive pooling therefore penalizes such articles, and coverage-aware indexing or fusion warrants evaluation.

Several limitations remain. The queries were automatically extracted, cleaned, and deduplicated from dictyBase pages using rule-based procedures rather than being rewritten or reviewed in detail by domain experts. Some compact claims may therefore depend on surrounding page or paragraph context and be less self-contained than manually written search queries. In addition, inline citations provide useful retrieval targets but not exhaustive or definitive relevance judgments: other relevant articles may exist but remain unlabeled, and some difficult cases may reflect claim–citation mismatch rather than retrieval failure. For example, a citation may support the surrounding note rather than the extracted sentence, may depend on context lost during extraction, or may simply be an imperfect curator choice. The benchmark is therefore best interpreted as a cited-article retrieval task rather than a complete relevance benchmark. Finally, the evidence-level labels are LLM-assisted analysis labels, not definitive expert annotations. Future work should extend the benchmark with expert-reviewed queries, broader model-organism coverage, and evidence-level evaluation of generated answers.

Appendix

Implementation details

BM25 retrieval uses PyTerrier [14] over a Terrier index with the default BM25 weighting model. Dense retrieval uses abhinand/MedEmbed-small-v0.1 [1] to encode queries and documents, with cosine similarity over an HNSW index. Retrieval fusion uses RRF with $k = 60$ and equal weights for BM25 and dense retrieval. Cross-encoder rerankers receive the top 1,000 fused candidates and are run with `max_length=512`. Final rankings are produced by post-rerank fusion: RRF combines the reranker top 200 with the retrieval-fusion top 200 using weights 0.8 and 0.2, respectively. These settings keep the final ranking primarily reranker-driven while preserving part of the retrieval-stage signal. Evidence-level labels are produced once using a fixed Llama-3.3 prompt schema [8], included in the repository, and frozen for all downstream analyses. Code, workflow configurations, and the public dataset (released as the dictyBase citation goldset)

are available at <https://github.com/fulaibaowang/dictycite>; the repository also includes the reusable retrieval/reranking pipeline used in the experiments. The Ragnarok comparison uses code from <https://github.com/castorini/ragnarok>.

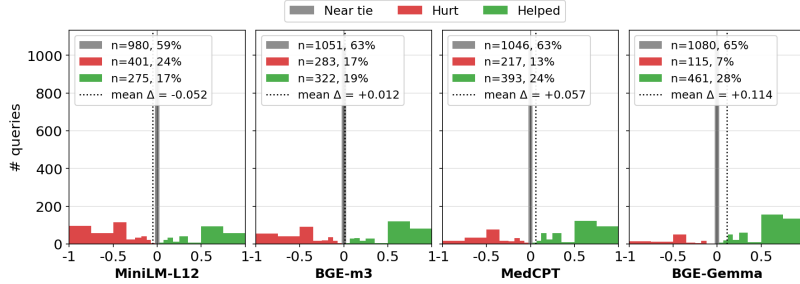


Fig. S1. Per-query reranking effect relative to BM25. $\Delta\text{MRR}@10$ is computed as reranker minus BM25 and grouped as hurt, helped, or near tie using thresholds of ± 0.025 . Similar aggregate scores mask different query-level behavior: MiniLM-L12 is mostly harmful, BGE-m3 is balanced, and MedCPT and BGE-Gemma shift more queries toward positive deltas.

Table S1. Effect of gene-aware query expansion on the query-expansion subset. Each cell reports $\text{MRR}@10$. For expanded-query variants, parentheses report the difference from the original-query baseline for the same reranker, in percentage points, with 95% paired-bootstrap confidence intervals.

Reranker	Original	+ Synonyms	+ Synonyms and products
BGE-m3	0.576	0.623 (+4.7 [+2.6,+6.8])	0.622 (+4.6 [+2.2,+7.1])
MedCPT	0.624	0.682 (+5.8 [+3.7,+7.9])	0.673 (+5.0 [+2.7,+7.2])
BGE-Gemma	0.694	0.703 (+0.9 [-0.5,+2.3]) [†]	0.701 (+0.7 [-0.6,+2.1]) [†]

[†] Confidence interval crosses zero. Deltas are in percentage points.

Table S2. Recall@1000 and BGE-m3 $\text{MRR}@10$ for abstract-only (Abs) versus abstract-plus-chunks (+Chk) retrieval, by abstract-evidence stratum and full-text coverage. Within each stratum, claim-stratum entries are split by coverage—has-PDF if at least one cited article carrying that label has chunked full text, and no-PDF otherwise—and scored on the same all-query runs against the cited articles carrying that label. In each +Chk column, parentheses give the change from Abs (Δ , percentage points) with 95% paired-bootstrap confidence intervals. Stratum sizes are given in Table 1.

Evidence stratum	R@1000		MRR@10	
	Abs	+Chk (Δ [95% CI])	Abs	+Chk (Δ [95% CI])
<i>has-PDF entries</i>				
<i>abstract supports detail</i>	0.993	0.997 (+0.4 [-0.1,+1.1])	0.802	0.843 (+4.2 [+2.3,+6.2])
<i>abstract supports core</i>	0.986	0.998 (+1.2 [+0.5,+2.1])	0.505	0.636 (+13.1 [+10.2,+15.9])
<i>abstract insufficient</i>	0.784	0.972 (+18.8 [+14.8,+23.0])	0.161	0.385 (+22.5 [+18.5,+26.4])
<i>no-PDF entries</i>				
<i>abstract supports detail</i>	1.000	1.000 (0.0)	0.815	0.731 (-8.4 [-12.6,-4.6])
<i>abstract supports core</i>	0.957	0.957 (0.0)	0.431	0.332 (-9.9 [-14.4,-5.7])
<i>abstract insufficient</i>	0.838	0.757 (-8.1 [-18.9,+0.0])	0.170	0.086 (-8.4 [-14.5,-3.3])

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